

Additional cases of *SKOR2* variants with cerebellar hypoplasia, a spectrum of developmental delays, and additional clinical characteristics.

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Abstract:

Cerebellar hypoplasia (CH) is defined by decreased cerebellar and/or vermis size. CH can be caused by multiple etiologies, such as prenatal infections, teratogenic exposures, metabolic conditions, chromosomal abnormalities, or single gene variants. *SKOR2* was initially studied in mouse models and was found to be involved in cerebellar development, *Skor2*-null mice cerebellum lacked normal cerebellar development, had smaller cerebellum, abnormal foliation and reduced size of the central vermis. Valence and colleagues previously reported one patient with homozygous null variant in *SKOR2* (p.Ser917*) and had developmental delays, ataxia, and cerebellar dysplasia. We here report four additional cases with *SKOR2* variants, and provide clinical description with brain imaging findings. Further studies may be needed to provide evidence for causation of *SKOR2* variants and cerebellar abnormalities. *SKOR2* gene has not been known as a human disease-causing gene thus far, but can be a good candidate.

Background:

Cerebellar hypoplasia (CH) is a descriptive term based on neuroimaging (Poretti et al., 2014), and is defined by decreased cerebellar volume, decreased cerebellar vermis size on midsagittal section, with (near) normal shape (Poretti et al., 2009). Due to the reduction of cerebellar size, the subarachnoid space is enlarged (Poretti et al.,

2009). Hypoplastic cerebella may result from decreased production of cells, increased apoptosis of cerebellum, or premature apoptosis or abnormal cell migration of cells (Patel & Barkovich, 2002). Cerebellar atrophy (CA) is different from CH, and sometimes difficult to differentiate even with brain imaging (Poretti et al., 2015). Patel and Barkovich used the term CA when the cerebellum is small, with large cerebellar fissures and shrunken folia, or if it underwent progressive volume loss; while CH was used to describe small cerebellum with normal size fissures compared with the folia (Patel & Barkovich, 2002). It is important to differentiate between CA and CH for management, prognosis and diagnostic work up purposes (Poretti et al., 2014).

The cerebellum develops early in embryonic life, and continues to develop into the first postnatal years (ten Donkelaar et al., 2003). CH can develop due to heterogeneous etiologies, such as congenital infections (Steinlin et al., 1996) (Melo et al., 2016), maternal smoking during pregnancy (Ekblad et al., 2010), prenatal alcohol exposure (Norman et al., 2009), teratogenic exposures during pregnancy such as some antiepileptic medications (Squier et al., 1990) (Main & Kulesza, 2017), metabolic disorders such as congenital disorder of glycosylation type 1a due to pathogenic/likely pathogenic biallelic variants in *PMM2* (Sparks & Krasnewich, 1993), or due to genetic conditions with single genetic variants or chromosomal abnormalities (Poretti et al., 2014) (Zou et al., 2018). CH can result from malformations (non-progressive congenital anomaly of an organ due to a primary development alteration), or disruptions (non-progressive congenital breakdown of the anatomy of an organ that could potentially have had a normal development) (Hennekam et al., 2013). Furthermore, CH can have general involvement of the entire cerebellum, or less often involvement of a part of the

cerebellum affecting the vermis or one/both cerebellar hemisphere without affecting the vermis (Poretti et al., 2014).

Studies have looked into genetic factors that aid in early development and organization of cerebellum such as *Wnt* family of signaling molecules (Rodriguez & Dymecki, 2000) (Subashini et al., 2017), and others found the expression of En-2 of the murine genes being expressed in the granular layer of cerebellum in mouse models (Davis & Joyner, 1988). In addition, other studies found genetic alterations in patients with CH, some appear syndromic, and some are non-syndromic. Tripathy and colleagues reported that mutations in *MAST1* gene leads to mega-corpus-callosum syndrome, with CH and cortical malformations (Tripathy et al., 2018). Missense *do novo* variants in *WDR37* have been identified in patients with CH, colobomas, epilepsy, intellectual disability and developmental delays (Kanca et al., 2019). Furthermore, chromosomal copy number variations were found in prenatal cases who carried fetuses with CH, where 5p15 deletion, 6q terminal deletion, and aberrations of X chromosome were the most common chromosomal abnormalities (Zou et al., 2018).

We here report four patients with *SKOR2* variants, cerebellar hypoplasia and developmental delays. We present their facial features, clinical and developmental achievements, and their brain imaging findings, and compare with one other previously reported case with *SKOR2* variants (Valence et al., 2019). *SKOR2* gene has not been known as a human disease-causing gene thus far, but can be a good candidate.

Case Reports:

Patient 1:

Patient 1 was initially evaluated by Genetics clinic at age 16 months due to cerebellar hypoplasia and developmental delays. His parents became concerned at 6 months of age when he did not have head support and was not rolling over. He was noted by the parents that overall he gets sick a lot, and that he tires easily. At 5 years of age, modified barium swallow revealed aspiration to liquids, and the family were instructed to thicken liquids to moderate thickness consistency. Developmentally, he was rolling from side to side at 7 months of age, he had head control at 10 months of age. At 16 months, he was able to transfer objects from hand to hand, and to finger feed himself. He was cooing and babbling but had no syllable sounds at 16 months of age. He walked at age 3 years. His gait was wobbly and had poor balance.

He was born to a 24 year-old mother, G2P2, and 41 year-old father. He was born after a full-term pregnancy, 38+2/7 weeks. He was born via vaginal delivery. His birth weight was 6 pounds (11th percentile for gestational age). No prenatal exposures to medications, toxins, alcohol, tobacco. No known prenatal infections. In terms of family history, he had a sister who is 2.5 years older than him, she is in good health. Both of his parents are in good health. No history of developmental delays or congenital anomalies on either side of his family. Both parents are Moroccan. The parents are second cousins once removed (maternal grandmother's father is the brother of paternal grandmother).

On physical exam, patient 1 had relative macrocephaly, dolichocephaly, depressed nasal tip, hypoplastic short nose, depressed nasal bridge, smooth philtrum, epicanthal folds, normal set ears, high arched palate, erythematous area on forehead. He had one cafe au lait spot (2 cm in diameter) on lower back. He had clinodactyly of

right hand 4th and 5th fingers, persistent fetal pads on both hands; flat feet, rocker bottom. Brain MRI at 9 months of age revealed inferior vermian hypoplasia, bilateral cerebellar hemisphere hypoplasia, small pons/brainstem, and somewhat elongated corpus callosum.

Genetic testing: Chromosomal microarray in 2015 was negative, and had revealed more than 3% regions of homozygosity. Pontocerebellar hypoplasia 18-gene panel (*CASK*, *CHMP1A*, *CLP1*, *EXOSC3*, *OPHN1*, *RARS2*, *RELN*, *SEPSECS*, *TSEN2*, *TSEN34*, *TSEN54*, *TUBA1A*, *TUBA8*, *TUBB2B*, *TUBB3*, *VLDLR*, *VPS53*, *VRK1*) sequencing and del/dup (at GeneDx in 2016) did not identify any pathogenic variants or variants of uncertain significance. Initially, whole exome sequencing (WES) (at Claritas in 2016) only revealed a variant of uncertain significance (VUS) in *NGLY1* (c.1804T>C) that was inherited from the father, no second variant identified in this gene. *NGLY1* is an autosomal recessive gene that is known to be associated with congenital disorder of deglycosylation (OMIM 615273). In 2019, he had WES at GeneDx which revealed a *de novo* VUS in *HDAC3* gene NM_003883.3 (c.916T>C; p.Cys306Arg) at 16% mosaicism; as well as Homozygous VUS in *SKOR2* gene NM_001278063.1 (c.1877delC; p.Pro626Glnfsx156), maternally and paternally inherited.

Patient 2 (from Allan):

Patient 2 is a 14-year-old female, the first child of non-consanguineous healthy Turkish parents, was born at 40 weeks of gestation after a normal pregnancy. Birth weight was 3250gram while data on birth length and the head circumference are not

available. Apgar scores are unknown. There were no concerns following the birth and the patient was discharged.

At six months of age, she showed normal development and could smile and babble, and was able to sit briefly with support. However, following the onset of epileptic seizures at the age of seven months, her development regressed and today she suffers from a severe intellectual disability and developmental delay. She is unable to walk or talk but understands a few words.

The epilepsy semiology is a mixture of both focal hypo- and hypermotor (hemiclonic) seizures with impaired awareness sometimes evolving to bilateral tonic clonic seizures. Repeated EEGs show a slowing of the background activity in the occipital lobes and multifocal interictal epileptic discharges also in the occipital lobe. Despite a combination of several antiepileptic drugs she remains refractory and has several seizures weekly.

Her physical exam reveals some dysmorphic features including: a scare face, thick eyebrows, telecanthi, upslanting palpebral fissures, ptosis, long eyelashes, a narrow nasal bridge with a normal set columella, short and deep philtrum, full cheeks and lips and small earlobes. Most recent exam at age 14, she has a scoliosis and contractures of the lower extremities. A brain MRI performed at the age of 1 and 9 years showed a small pons and vermis with enlargement of interfolium spaces (anterior lobe) and an enlarged primary fissure.

Genetic testing included a next generation sequencing (NGS) panel performed at the age of 14 years revealed a *de novo* inframe deletion in the *SKOR2* gene NM_001278063.1: (c.1317_1331delAGGCGGCGCGGGCGC; p.Gly440_Ala444del).

Patient 3 and 4 from Silke and Rami:

Patient 3 and patient 4 are siblings, the fifth and second children of consanguineous, healthy parents who live in Germany, originating from Iraq (first degree cousins). Maternal family history was notable for a late abortion (during 8th month gestation, multiple malformations, no medical reports from Iraq where late abortion had happened) and a nephew with ID and a complex heart failure (no medical reports, the family lives in Iraq). Karyotyping, chromosomal microarray analysis and targeted testing for Fragile-X revealed normal results in both siblings.

Patient 3:

Patient 3 was the fifth child of healthy consanguineous parents from Iraq (first degree cousins). He was born at 41 weeks of gestation by cesarean section with a birth weight of 4380 g (+1.43 SD), a length of 53 cm (- 0.04 SD) and a head circumference of 36 cm (+ 0.08 SD). Immediately after birth, he was admitted to hospital due to tetralogy of Fallot with pulmonary stenosis (surgically corrected). He presented with congenital cataract, hyperopic astigmatism (right eye: + 7,50 dpt., left eye: + 8,50 dpt.), bilateral sensorineural hearing impairment, small right kidney, kidney and ureteral duplication left. He displayed severe psychomotor developmental delay and intellectual disability (ID). He was able to walk without support at age 24 months. Until the age of 10 years, he was solely able to walk short distances (100 to 200 meters) and was dependent on a wheelchair (the reason for it is unclear). Currently he is able to walk distances of about 1,000 meters. Speech was nearly absent. Aged 17 ¹/₁₂ years, he spoke few single

words. He had severe ID (IQ: 58). Electroencephalography (EEG) was normal. He did not display febrile or non-febrile seizures. MRI scans during a period of about 4 years (between ages 4 and 8 years) showed nonspecific non-progressive frontal and occipital white matter lesions and leukoencephalopathy. He had deficits in social communication and interactions with the hallmarks: i) anxiety-related behavior; ii) low levels of frustration tolerance; iii) distress at small changes; iv) tendency to social withdrawal; v) impaired social interaction, lack of peer relationships and; vi) sleep-wake cycle disturbance, trouble falling asleep, and daytime somnolence.

On physical examination at age 17 $\frac{1}{12}$ years, weight (56 kg, – 1.48 SD) and OFC (55 cm, – 1.15 SD) were in the normal range, whereas height (158 cm, – 3.06 SD). He had a slender habitus, and minor facial aspects including a long face, a myopathic facies, thick eyebrows, telecanthi, hypertelorism, downslanted long palpebral fissures, narrow nasal bridge, broad nasal tip, short columella, short and smooth philtrum, a thin upper lip, bilateral sandal gaps and flat feet.

Patient 4:

Patient 4 was the second child of healthy consanguineous parents from Iraq (first degree cousins). She was born spontaneously after an uneventful pregnancy. ?Week of gestation and birth measurements were not remembered but reported to be normal. It was a home birth in Iraq. She displayed hypotonia and anal atresia (surgically corrected in Iraq). Milestones of motor and speech development were delayed. She was able to walk without support at age 24 months. She spoke first words at age of 5 years. Aged 29 $\frac{9}{12}$ years, she spoke few simple sentences. She also had severe ID (IQ: 55). She

displayed bilateral nystagmus and bilateral hyperopic astigmatism (right eye: + 5,60 dpt., left eye: + 6,30 dpt.). Electroencephalography (EEG) was normal. She did not display febrile or non-febrile seizures. Brain MRI scans have not been performed.

She had a friendly behavior but mild deficits in social communication and interactions with the hallmarks impaired social interaction and lack of peer relationships. On physical examination at age 29 ⁹/₁₂ years, weight (63 kg) and OFC (56 cm) were in the normal range, whereas height (158 cm) was below the normal for age. She presented with craniofacial dysmorphisms. She had minor facial aspects including a long face, thick eyebrows, telecanthus, hypertelorism, downslanting long palpebral fissures, broad nasal tip, short and smooth philtrum, a thin upper lip, low-set ears, bilateral sandal gaps and flat feet.

Methods:- Discussion:

SKOR2 gene and protein. SKOR2 gene (in the older literature sometimes also referred as *CORL2* or *Fussel-18*), encoded on the chromosome 18, has been shown to express during both embryogenesis and adult life in the cerebellum, the spinal cord, dorsal root ganglia and at very low levels in the testes (Arndt et al., 2005). In mice SKOR2 is expressed in Purkinje cells where it physically interacts with SMAD proteins to act as a transcriptional regulator of SHH expression. SHH signalling between Purkinje and granule cells is essential for the correct development of cerebellum, the central motor coordination unit in the hindbrain of all vertebrates (Wang et al., 2011). Cerebellar anomalies lead to disorders in movement, equilibrium, posture and motor

development (Fine et al., 2002). Recently, a homozygous null variant in *SKOR2* was reported in a child with cerebellar ataxia and cerebellar hypoplasia (Valence et al., 2019).

SKOR2 has only five notable paralog genes – *SKOR1*, *SKI*, *SKIL* and *DACH1* and *DACH2* though none of them is too closely related to *SKOR2*, and the homology is deduced through the similarity in several domains they share. All paralogs share the Dachshund domain, while the *SKOR* and *SKI* families share the SMAD-interacting domain (Fig 1). The sequence similarity in these regions (Mihalek et al., 2004) (Zhang et al., 2013) across distant paralogs, as well as strong conservation across *SKOR2* in all vertebrates, ~~confirms the ordered protein regions, suggest they represent ordered domains of protein structure. which indeed have~~ This has indeed been confirmed by crystallography (Wilson et al., 2004) (Wu et al., 2002) (Fig 1). The intervening peptide sequences probably function as disordered linkers, providing the re-organizational flexibility for the complexes that *SKOR2* engages-tolerating more variation in the amino acid content and the peptide length.

The molecular impact of *SKOR2* mutations reported in this work. The homozygous *SKOR2* frameshift mutation in patient 1 results in loss of a functional transcript through nonsense mediated decay. Similarly, in Patients 3 and 4, the protein product is lost, but through a different mechanism. There, both alleles carry a substitution in the first position in one of the donor splice regions, the guanine crucial for the two-step biochemistry of the splicing process (Sperling et al., 2008) [11,12 can't find ref,13]. Thus the process never results in a translatable mature mRNA – with an exon

skipped or included, the product most likely results (again) in an early stop codon. The systems-level impact is the same as in the case of the Patient 1: no protein product from either copy of the *SKOR2* gene. The absence of *SKOR2* interrupts the signaling pathway(s) required for proper cerebellar development and organization which leads to the observed phenotype.

Only one mutation was identified in patient 2, though from the genotypes of the healthy people collected in gnomAD v2.1.1 database (Karczewski et al., 2020), we know that people with loss of one copy of the gene (through frameshift or missense mutation) can reach adulthood without any neurological symptoms. 15 such cases have been reported in gnomAD, indicating strongly that *SKOR2* exerts its effect in a recessive way.

Furthermore, the one reported variant is an in-frame deletion in the coding sequence of the gene, resulting in deletion of 5 amino acids (p.Gly440_Ala444del) in what appears to be a disordered linker between functional, structurally ordered, domains in the *SKOR2* protein. This region is not strongly conserved, not even among mammals, suggesting that it is not critical for the protein function. The protein thus may express, fold, and even function to a certain degree. This would need to be validated experimentally. Coupled with the fact that no molecular lesion has been reported in the other gene we would like to draw attention to the possibility that the Patient 2 has been misdiagnosed.

We would like to note here that the *SKOR2* gene and its products are still a relative unknown. Perhaps a damaged but expressed mutant as we see in the Patient 2 engages in nonproductive (homo)dimerization, or otherwise competes non-productively

for the same resources as its healthy counterpart from the other copy of the chromosome, and thus exerts its pathological effect. We might have to revisit that possibility in the future after a better molecular characterization of *SKOR2* becomes available.

Discussion:

The SKOR2 phenotype. *SKOR2* gene was first identified in 2005 by Arndt, as one of Ski/Sno family of transcriptional co-repressors, where it was referred to as Fussel-18 (functional Smad suppressing element on chromosome 18) (Valence et al., 2019). Arndt et al., found that the expression of Fussel-18 is highly specific in neuronal tissues, especially in the cerebellum, the spinal cord, and dorsal root ganglia during embryogenesis and in adulthood, which was not the case for other Ski and Sno that are expressed ubiquitously in human tissues (Arndt et al., 2005). Subsequently, Minaki et al., 2018 confirmed that *Skor2* (authors referred to *Corl2*) expression was restricted to the cerebellum, ventral metencephalon, and myelencephalon of both embryonic and adult mice stages (Minaki et al., 2008). In addition, Wang et al., 2011 produced a *Skor2*-null mice cerebellum at various developmental stages (Wang et al., 2011). They found that the *Skor2*-null mice cerebellum lacked the four principal fissures observed in the control cerebellum, smaller cerebellum, abnormal foliation and severely reduced size of the central vermis in adult mice. In addition, the cerebellar cortex was found to be thinner compared with the wild-type cerebellum (Wang et al., 2011). Wang et al., concluded that loss of *Skor2* leads to defective Purkinje cell development, and that *Skor2* is required for proper expression of *Shh* (Wang et al., 2011). Similarly, another

study confirmed the small cerebellum and the less developed foliation in the *Skor2* knock-out mice (referred to *Corl2* in that study) (Nakatani et al., 2014).

The first human report of *SKOR2* variants in humans was published in 2019 by Valence and colleagues, as the authors studied exome sequencing in 20 patients with congenital ataxia (Valence et al., 2019). The patient reported in Valence et al., had homozygous null variants in *SKOR2* (NM_001278063.1:c.2750C>G; p.Ser917*). She is 10 years of age, born to a consanguineous Turkish parent, with neonatal hypotonia and psychomotor delay. She walked independently at age 2 years. She had scoliosis, nystagmus, mild ID, and nonprogressive ataxia. Her brain MRI revealed a unique pattern of cerebellar dysplasia, with predominance of the abnormal foliation in vermal lobules IV to VIII, as well as thick corpus callosum, a small posterior fossa, and thin pons.

Overall, CH can result in various presentations, with broad spectrum (Bolduc & Limperopoulos, 2009) (Wassmer et al., 2003). Over half of patients with CH have cognitive impairment or developmental delays (Wassmer et al., 2003). Language delays were described in more than half of patients with CH (Wassmer et al., 2003) (Ventura et al., 2006), the level of delays ranged from mild (76%) to severe or complete absence of language (19%) (Bolduc & Limperopoulos, 2009). Delayed motor milestones was described in 18-90% of children, autistic features in 5-20%, social and behavioral difficulties were reported in 71% of patients with CH (Bolduc & Limperopoulos, 2009). Neurological abnormalities were commonly described in patients with CH: ataxia and/or abnormal coordination (12-49%) (Ventura et al., 2006) (Wassmer et al., 2003), hypotonia (49-93%) (Ventura et al., 2006) (Wassmer et al., 2003), increased tone

(58%), and oculomotor disturbances in (35-57%) (Bolduc & Limperopoulos, 2009) (Wassmer et al., 2003). Microcephaly (Wassmer et al., 2003), nystagmus (Wassmer et al., 2003), and facial dysmorphism were also reported (Ventura et al., 2006). Mental retardation (MR) has also ranged from borderline, mild MR to severe MR (Ventura et al., 2006). In terms of prognosis, this may remain poorly defined, depending on the exact brain malformation within the cerebellar region (Pinchevsky et al., 2019). Severe global developmental delays were significantly associated with CH ($p=0.049$), whereas children with vermis hypoplasia were less likely to have severe global developmental delays ($p=0.003$) (Pinchevsky et al., 2019).

Conclusions:

We report four patients with *SKOR2* variants, two of which are siblings, who have cerebellar hypoplasia, various degrees of developmental and speech delays, with patient 2 being non-ambulant and non-verbal. We also compare the reported patients with one other previously reported patient. Previous animal studies provide evidence of abnormal cerebellar development in mice knock-out models (Pinchevsky et al., 2019) (Nakatani et al., 2014), after the gene was identified in humans (Arndt et al., 2005). *SKOR2* can be a good candidate for human disease-associations. More studies and additional patient reports are required to provide further evidence.

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Table 1: Clinical Phenotypes on Patients with *SKOR2* Variants

Clinical Characteristics	Patient 1 (BCH)	Patient 2 (Allan)	Patient 3 (Silke, Rami)	Patient 4 (Silke, Rami)	Published case (Valence 2019)	Total (4)
Date of birth	2014	2006	2002	1990		
Gender	M	Female	Male	Female	Female	
Age	5.5 years	14 years	17 years	29.9 years	10 years	
Ethnicity	Moroccan	Turkey	Iraqi	Iraqi	Turkish	
Consanguinity	Y		Y	Y	Y	
Current Country	USA	Denmark	Germany	Germany		
Gene: <i>SKOR2</i>	Homozygous variants	Heterozygous Inframe deletion (<i>De novo</i>)	Homozygous	Homozygous	Homozygous variants	
NM	NM_001278063.1	NM_001278063.1	NM_001278063.1	NM_001278063.1	NM_001278063	
Mutation (at DNA level) allele 1	c.1877delC	c.1317_1331 delAGGCGGCGGGGCGC	c.2752+1G>T	c.2752+1G>T	c.2750C>G	
Mutation (at protein level) allele 1	p.Pro626Glnfsx156	p.Gly440_Ala444del	p.?	p.?	p.Ser917*	
Putative effect	loss of protein	expressed dysfunctional	loss of protein	loss of protein	loss of protein	

		product	protein	product	product	product	
Mutation (at DNA level) allele 2	c.1877delC			c.2752+1G>T	c.2752+1G>T	c.2750C>G	
Mutation (at protein level) allele 2	p.Pro626Glnfsx156			p.?	p.?	p.Ser917*	
SKOR2 variant found by: - targeted analysis - gene panel - WES/WGS	WES	WES		WES with sibling	WES with sibling		
Mutation confirmed by Sanger sequencing?		Yes					
Other findings on WES	<i>HDAC3</i> c.916T>C (p.Cys306Arg) <i>de novo</i> VUS mosaic	No other pathogenic variants		Homozygous LoF variant in <i>PAN2</i> (NM_001127460.29): Chr12:56721179; c.887dup; p. (Tyr296*) Homozygous missense variant in <i>FMNL3</i> (NM_175736.4): Chr12:50042077; c.2575C>T; p. (Arg859Trp)	Homozygous LoF variant in <i>PAN2</i> (NM_001127460.29): Chr12:56721179; c.887dup; p. (Tyr296*) Homozygous missense variant in <i>FMNL3</i> (NM_175736.4): Chr12:50042077; c.2575C>T; p. (Arg859Trp)		
Description of the patient in a few lines including health problems, behavior and level of functioning	Global developmental delays, cerebellar hypoplasia, hypotonia, poor appetite, febrile seizures, sleep disturbance.	Severely handicapped, nonverbal and nonambulant. Malformation of cerebellum and ataxia. Epilepsy since age of 7 months.				Neonatal hypotonia, psychomotor delay, ataxic gait, mild ID, scoliosis, nystagmus.	
Facial and Distal Limb Morphology							

Face, Square HP_0000321 (FS)/ Flat (FF) HP_0012368		Square	Long face	Long face		
Sparse hair HP_0008070		No (hypertrichosis with full and low hairline extending into the forehead)	-	-		
Thick eyebrows HP_0000574		+	+	+		
Telecanthus (T) HP_0000506 / Epicanthus (E) HP_0000286	E	T	T	+		
Palpebral fissures Upslanted (U) HP_0000582 / Downslanted (D) HP_0000494		U	D	D		
Palpebral fissures, short HP_0012745 /long HP_0000637		Long	Long	Long		
Ptosis (P) HP_0000508 / Belphephimosis (B) HP_0000581		P	-	-		
Long eyelashes HP_0000527		+	-	-		
Depressed nasal bridge HP_0005280	+	+	-	-		
Depressed nasal ridge HP_0000457	+	NA	-	-		
Narrow nasal bridge HP_0000446		+	+	-		
Short nose HP_0003196	+	-	-	-		
Broad nasal tip HP_0000455		+	+	+		
Short columella HP_0002000		-	+	-		
Anteverted nares HP_0000463		-	-	-		
Full cheeks HP_0000293		+	-	-		
Philtrum Short HP_0000322 (S)/Long (L) HP_0000343 /Deep (D) HP_0002002 /smooth (SM) HP_0000319	SM	S + D	S/ SM	S, SM		
Everted upper lip vermillion HP_0010803		-	-	-		
Thin upper lip vermillion HP_0000219		Full lips, both upper and lower	+	+		
High palate HP_0000218	+	NA	-	-		
Reduced number of teeth HP_0009804		NA but has had teeth removed. We just don't	-	-		

		know why.				
Micro HP_0000347 /retrognathia HP_0000278		-	-	-		
Low-set ears (L) HP_0000369		NA	-	L		
Protruding ears HP_0000411		-	-	-		
Prominent inferior crus of antihelix HP_0011238		NA	-	-		
Overfolded helix HP_0000396		NA	-	-		
hypoplasia of earlobe HP_0009906		+	-	-		
Ulnar deviation of finger(s) HP_0001193		NA	-	-		
Tapered fingers HP_0001182		NA	-	-		
Clinodactyly of the fifth finger HP_0004209	+	NA	-	-		
Prominent digit pads HP_0011298	+	NA	-	-		
Sandal gap HP_0001852		NA	+	+		
Pes Planus HP_0001763	+	NA	+	+		
Rocker bottom foot HP_0001838	+	NA	-	-		
Other dysmorphic features			Myopathic facies, hypertelorism	hypertelorism		

Characteristics and Medical problems	Patient 1	Patient 2	Patient 3	Patient 4	Published patient	Total
Gestational age (weeks)	38	40	41	? full term		
Growth						
Birthweight in grams (SDS) (if not available please provide p values)	2700 gram	3250 gram	4380 gram (+1.43 SD)	No data		
Length at birth in cm (SDS)		NA	53 cm (- 0.04 SD)	No data		
OFC at birth in cm (SDS)		NA	36 cm (+ 0.08 SD)	No data		
Age at last measurements (yr)	5.5 years	13 years and 11 mths	17 1/12 years	29 9/12 years		
Height in cm at last measurement (SDS)	107 cm (33%)	138cm	158 cm, - 3.06 SD	158 cm		
Weight at last measurement in kg	17.9 kg (40%)	44,4 kg	56 kg, - 1.48 SD	63 kg		
BMI at last measurement	15.6 (56%)	34	22	25		
OFC at last measurement in cm (SDS)	52.8% (84%)	NA	55 cm, - 1.15 SD	56 cm		
Development and Behavior						
Intellectual Disability: Mild, Moderate, Severe	Mild	Severe	Severe, IQ 58	Severe, IQ 55	Mild	
Age at walking	Age 3	Never, non-ambulant, never learned to sit or walk	2 years	2 years	Age 2	
Age at first words		Non-verbal, understands a few words.	Nearly absent speech	5 years		
Autism/autism-like behavior		Too handicapped to assess	Yes, not formally tested deficits in social communication and interactions with the hallmarks: i) anxiety-related behavior; ii) low levels of frustration tolerance; iii) distress at small changes; iv) tendency to social withdrawal; v) impaired social interaction, lack of peer	Yes, not formally tested, mild deficits in social communication and interactions with the hallmarks impaired social interaction and lack of peer relationships		

Other behavioral problems		If yes, please describe	relationships sleep-wake cycle disturbance, trouble falling asleep, daytime somnolence	No		
Senses						
Nystagmus		Y	Y	bilateral nystagmus and bilateral hyperopic astigmatism	Y	
Esotropia	Y	NA	-	-		
Hypermetropia (Dpt)/ Myopia (Dpt). If yes, please describe how many Dpt		Dxt: -0,25 -2,50, 6 sinistre: -1,00 - 2,50, 16	hyperopic astigmatism (right eye: + 7,50 dpt., left eye: + 8,50 dpt.)	bilateral hyperopic astigmatism, right eye: + 5,60 dpt., left eye: 6,30 dpt.		
Strabismus		Y	+	+		
Cataract	No	No	-	n.a.		
Blindness	No	No	-	-		
Other problems with vision			-	-		
Hearing impairment	No	No	bilateral sensorineural hearing impairment	-		
MRI findings	malformed posterior fossa, including bilateral cerebellar hemisphere hypoplasia and inferior vermian hypoplasia with abnormal lobulation of the superior vermis, and a small brainstem.	T1-weighted sequence showing a small pons and vermis with enlargement of interfolium spaces (anterior lobe) and an enlarged primary fissure.	Bi-frontal unspecific white matter lesions, non-progressive during a period of 4 years, cerebellum and pons normal.	Not done	thick corpus callosum, a small posterior fossa, a thin pons, and a unique pattern of cerebellar dysplasia, with predominance of the abnormal foliation in vermian lobules IV to VIII	
Malformations and other health problems						
Cerebral anomaly						
Seizures	Febrile seizures	Yes onset at 7 months of age. Currently, she has an intractable epilepsy and a multifocal EEG.	no	no		
Cleft palate		No	-	-		

Laryngeal anomaly		No	-	-		
Cardiac anomaly		No	tetralogy of Fallot with pulmonary stenosis	-		
Gastro-intestinal anomaly		NA	-	Anal atresia		
Renal anomaly		NA	small right kidney, kidney and ureteral duplication left	-		
Cryptorchidism		No	-	-		
Hernia umbilical (U) inguinal(I)		No	-	-		
Scoliosis / kyphosis		Kyphosis of the thoracic spine	-	-	+	
Hip dysplasia	Clinodactyly of 4th and 5th fingers	NA	-	-		
Arm/hand anomaly	Flat feet, rocker bottom.	NA	-	n.a.		
Leg/foot anomaly		NA	-	-		
Hypermobility		Yes general	-	-		
Contractures		Yes	-	-		
Muscle hypertrophy/hypertonia		No	-	-		
Recurrent infections		NA	-	-		
Constipation		Yes	-	-		
Problems with intubation		NA	-	-		
Problems with certain medication		Yes urticarial rash due to lamotrigine and oxcarbazepine	-	+		
malignancies		No	-	-		
Keloids		NA	-	-		
Problems with teeth		She has probably had overlapping teeth, has had teeth removed. We don't have additional data on this on file	-	-		
Any hormonal disturbances (such as hypothyroidism, delayed puberty)		No	-	-		
Allergies		Yes urticarial	-	-		

		rash due to lamotrigine and oxcarbazepine				
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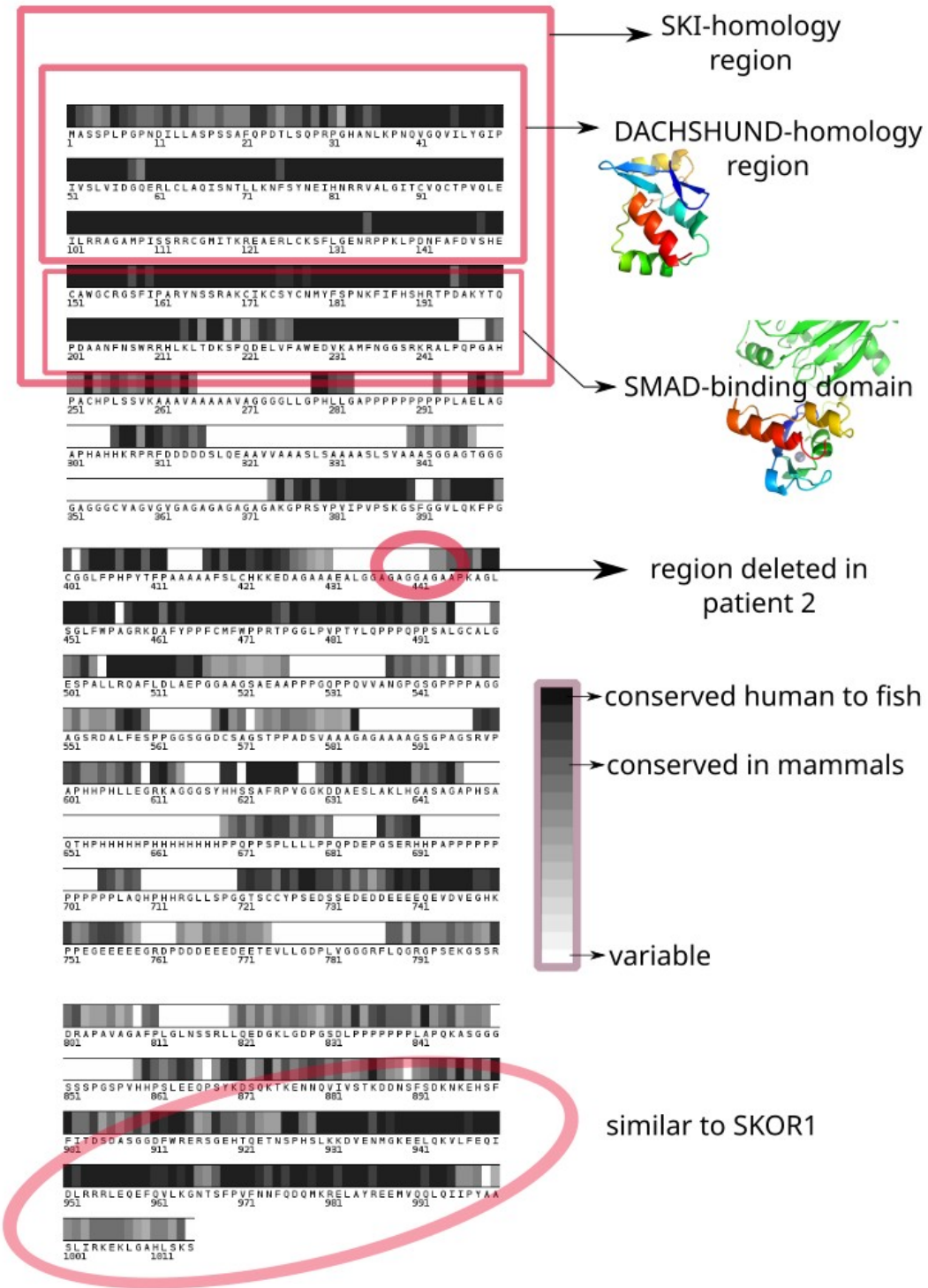


Figure 1: Domains and conservation on the SKOR2 sequence. More conserved regions are indicated by darker shades. They most likely correspond to structurally ordered

regions. For two of those regions the ordered protein structure has been confirmed crystallographically. Conversely, non-conserved regions commonly correspond to disordered regions on peptide sequence that better tolerate mutation. The deletion from Patient 2 belongs to one such region.

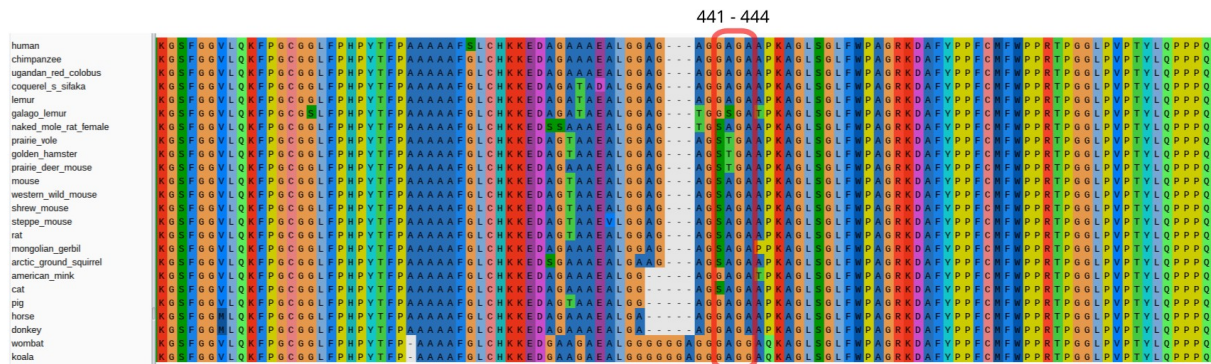


Figure 2: The alignment of SKOR2 orthologues in the vicinity of the position deleted in Patient 2. Notable is the variability in the composition and the length.